Advanced Computer Networks – PCS223

Assignment 5

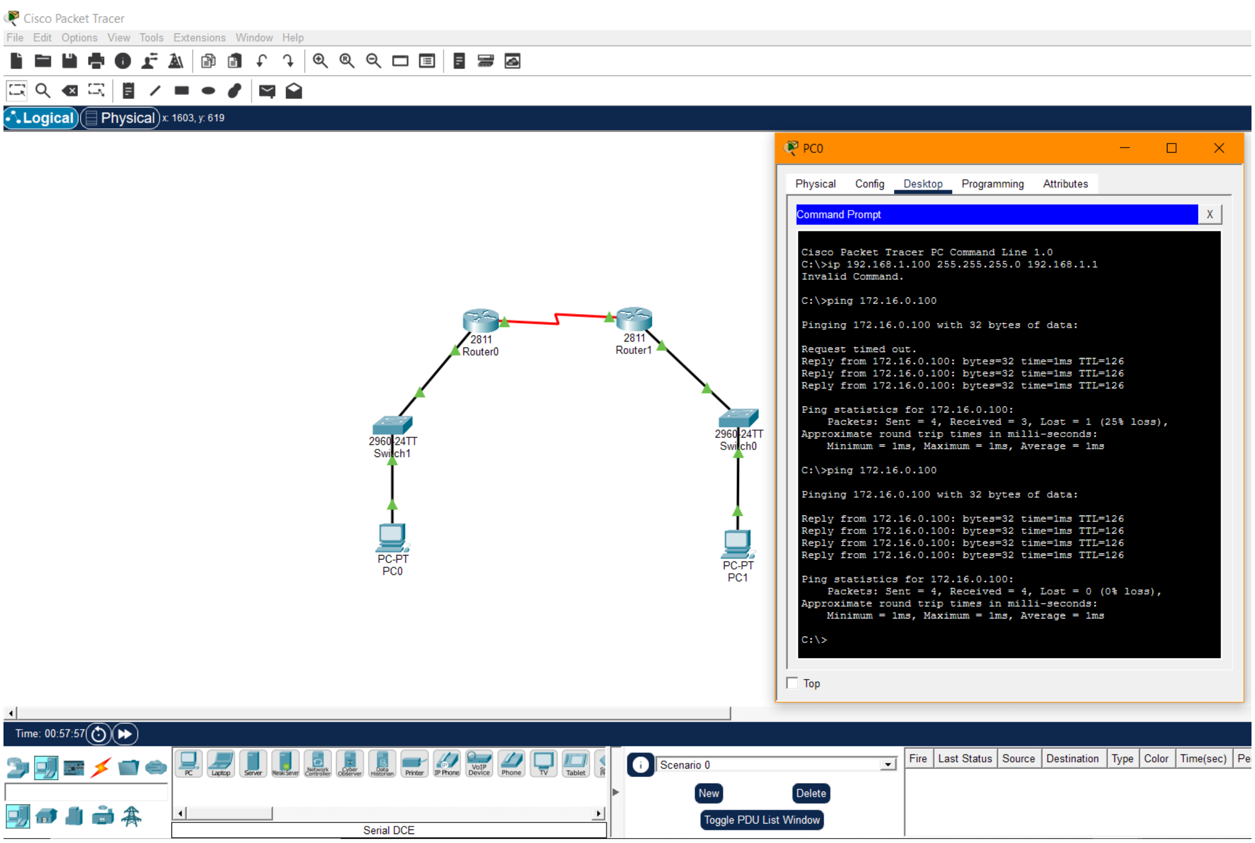
Name – Dhruv Garg
Subgroup – CS2
Roll No. – 8025320034

Q.1 Objective: Configure Static Route in Cisco packet tracer

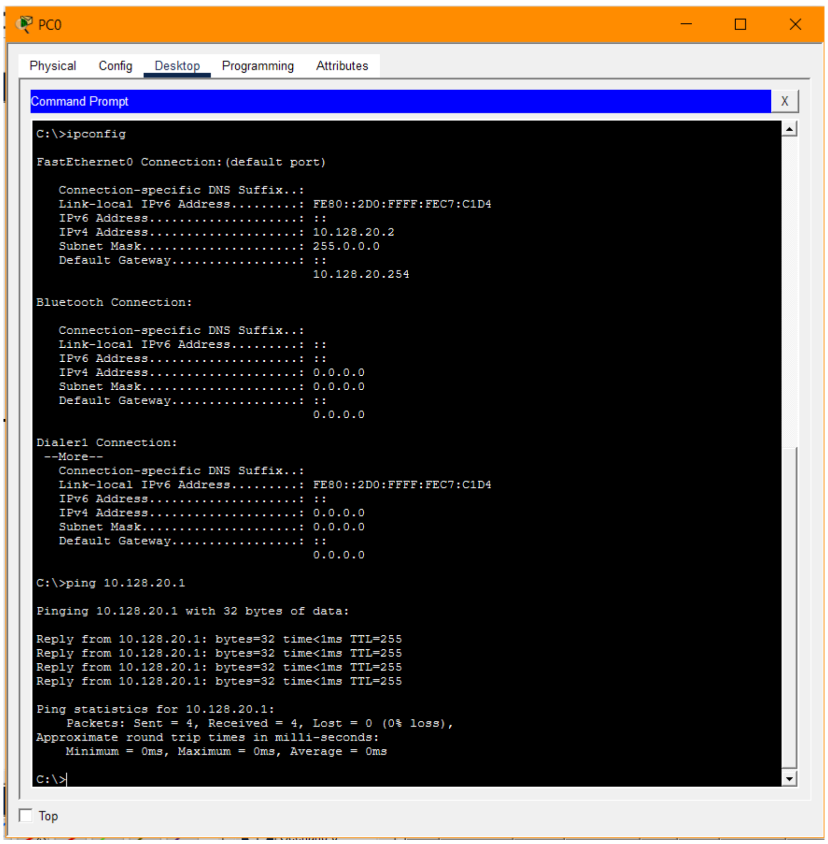
- Static route tells the device exactly where to send traffic, no matter what.
- Static route is often used when your network has only a few routers or there is only one route from a source to a destination.

Scenario

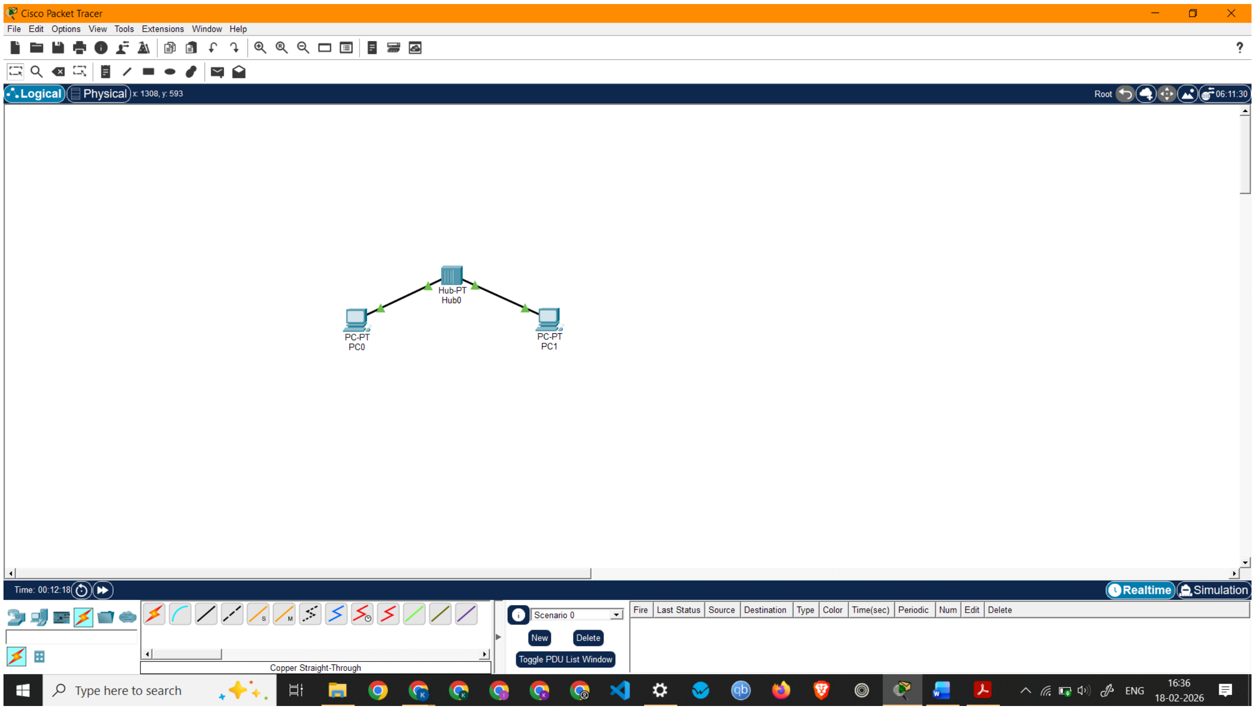
- Suppose that your company has 2 branches located in Tehran and Shiraz.
- As the administrator of the network, you are tasked to connect them so that employees in the two LANs can communicate with each other.
- After careful consideration you decided to connect them via static route.

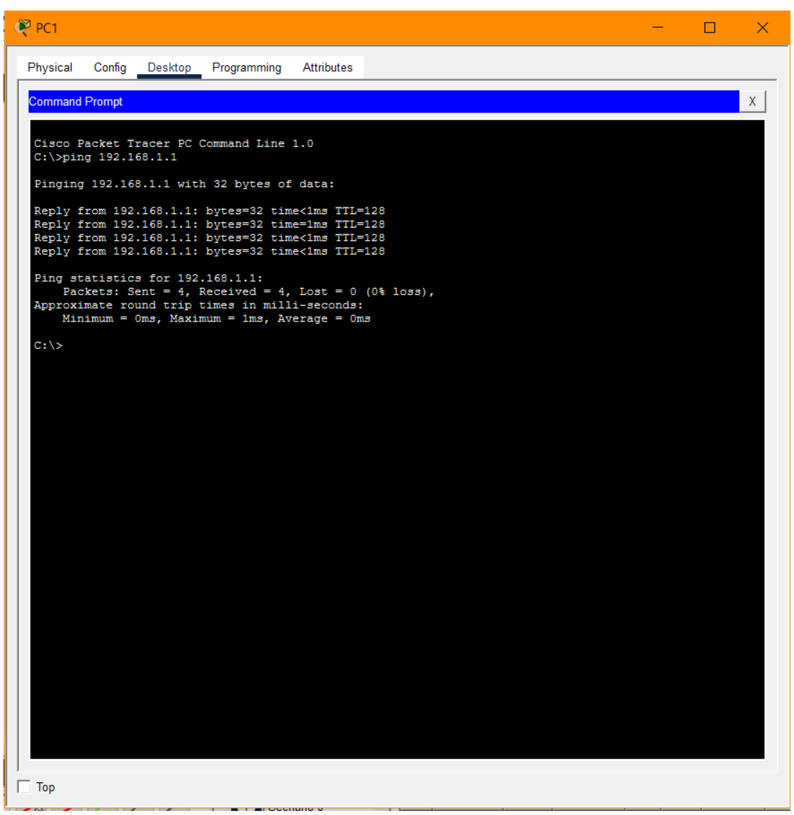


Q.2 Suppose you want to create 3 subnets of a network having network address 198.168.1.0. In another network with network address 220.168.2.0, you want to create 2 subnets. For this scenario configure the end devices and routers accordingly. Connect two different networks (both are subnetted) through static routing and show the message transmission among hosts of the two networks.

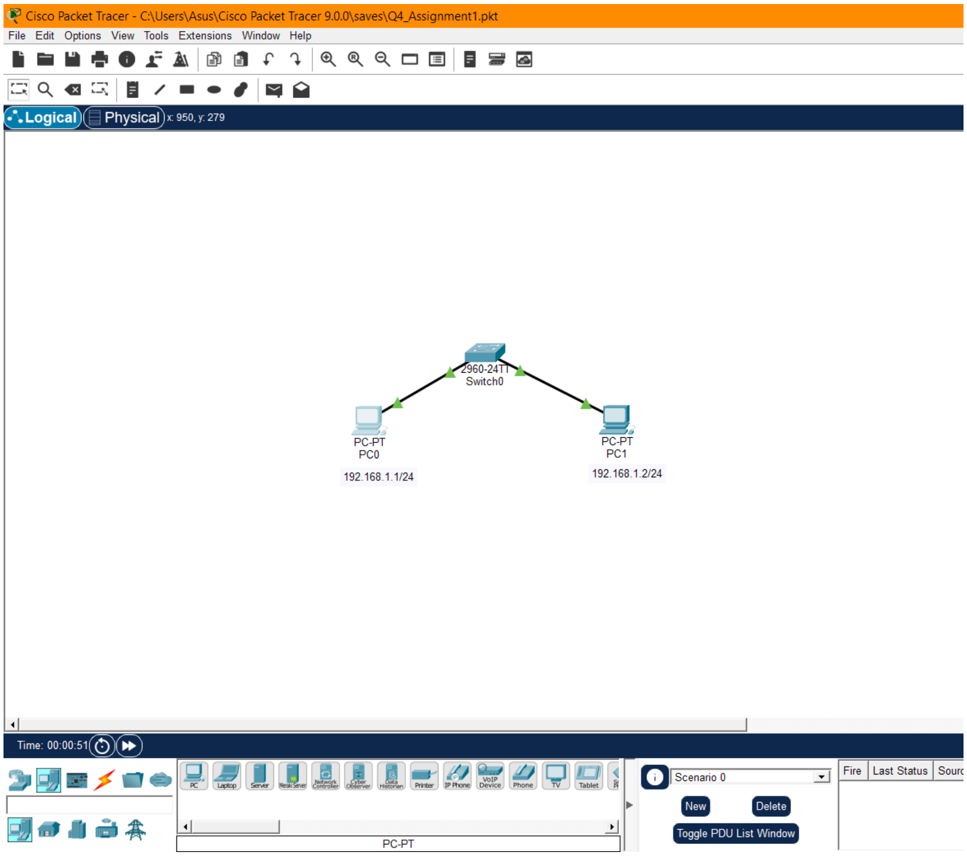


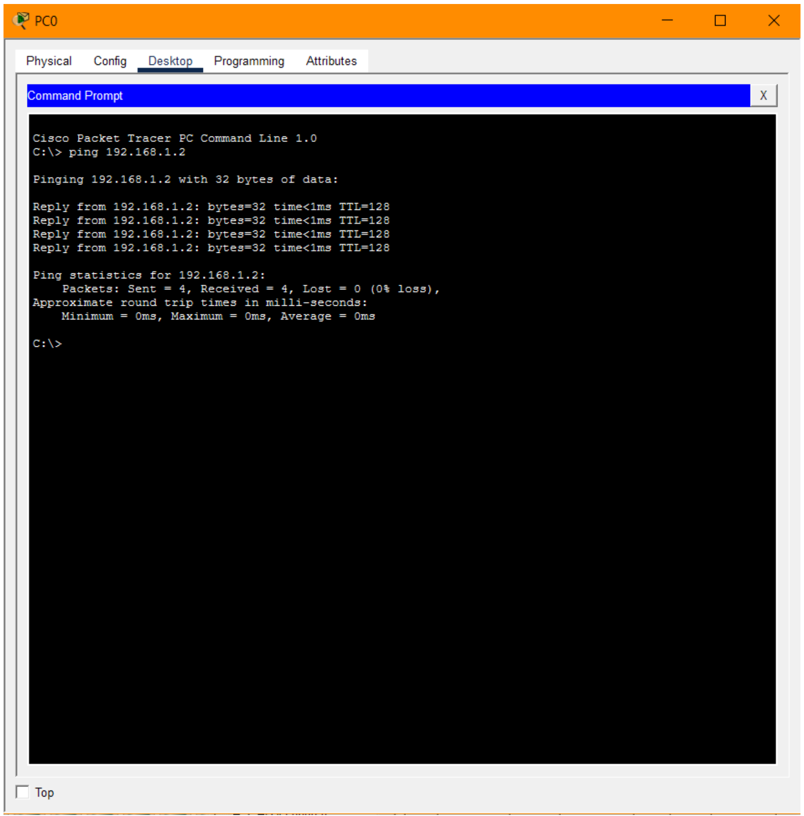
1. Create a network of 2 PCs as below.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Hub-PT
Hub0
PC-PT
PC0
PC-PT
PC1
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Copper Straight-Through
Type here to search

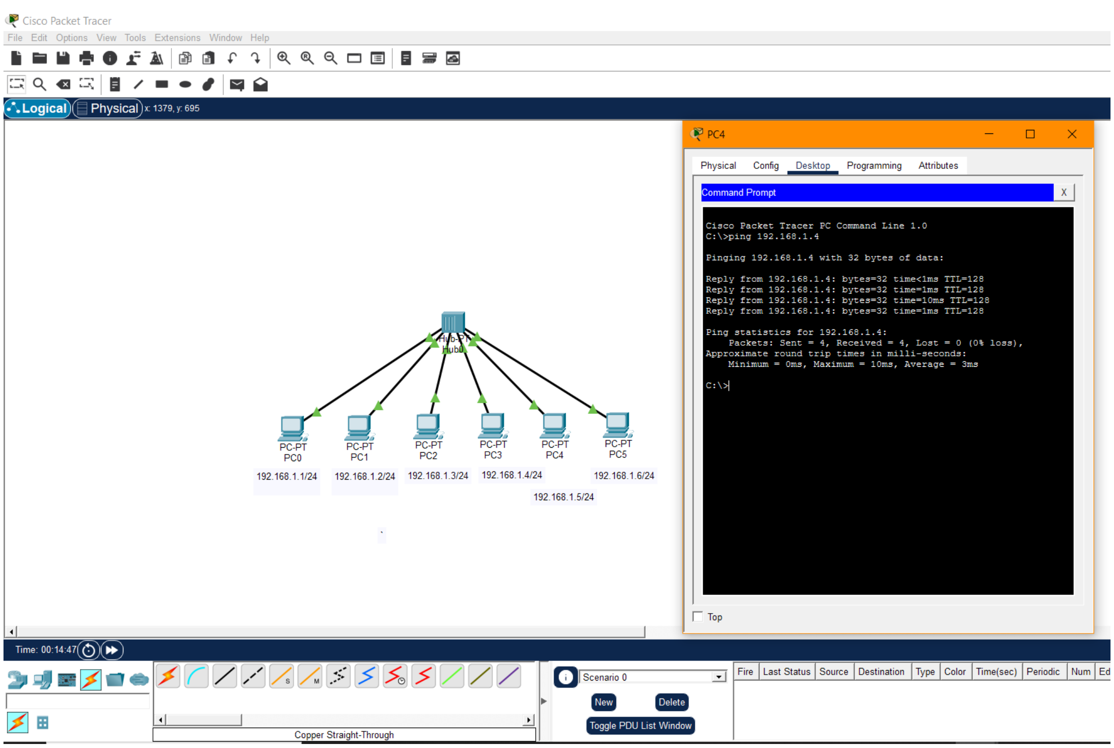

PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

2. Connect 2 PCs with a Switch below.

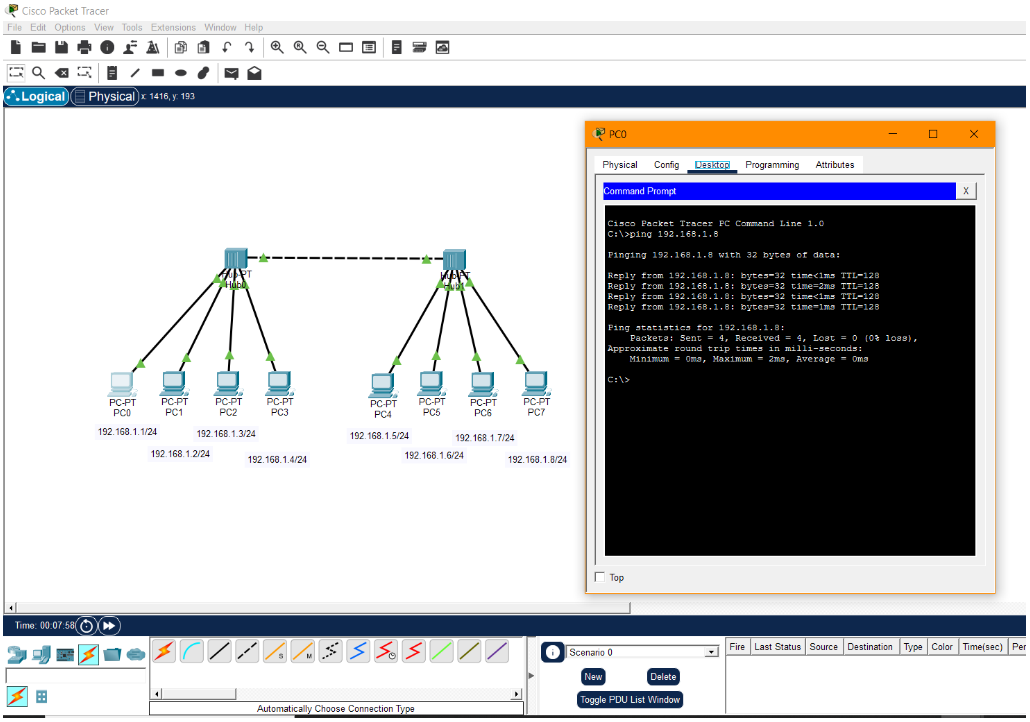




3. Create BUS topologies as below.

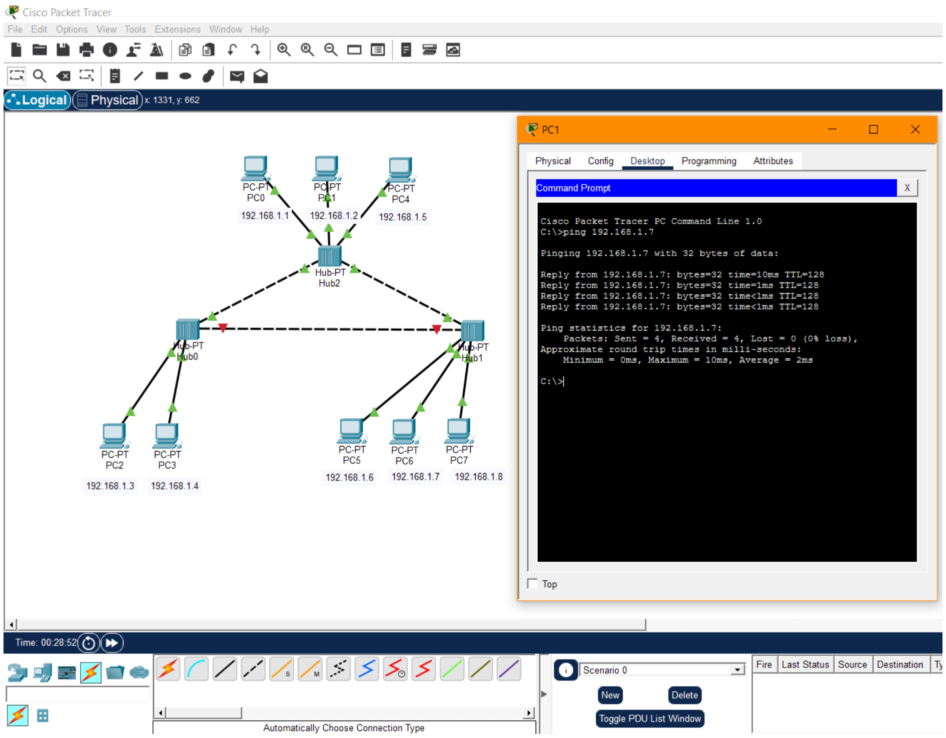


(a)

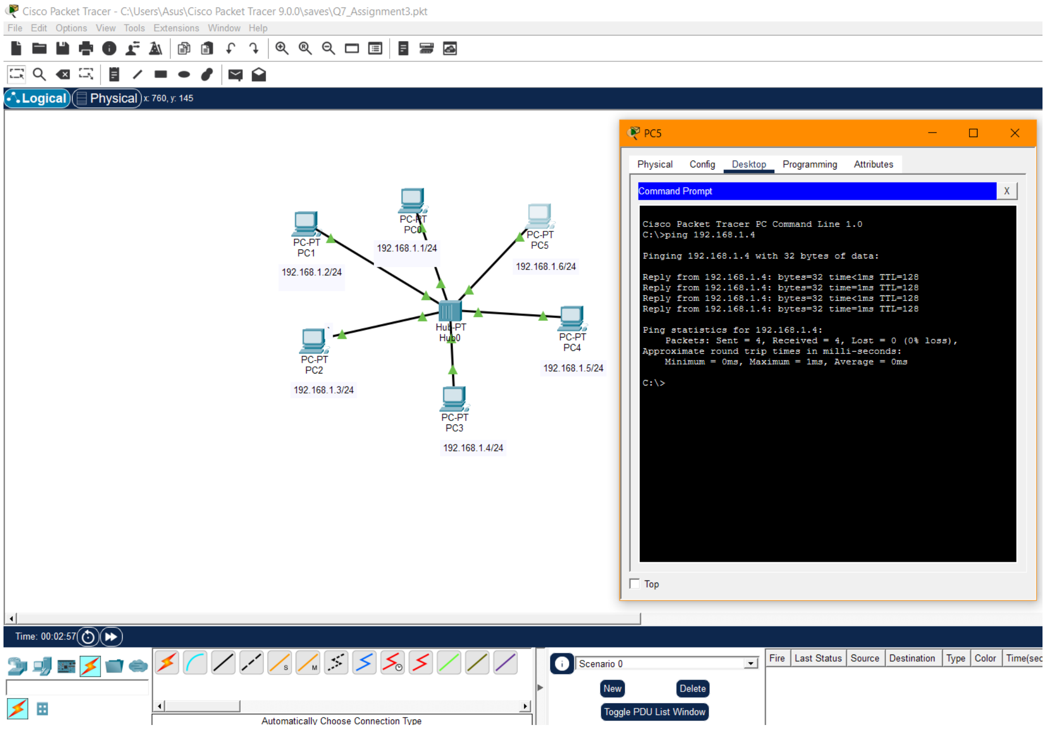

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1416, y: 193
Hub-PT
Hub0
Hub-PT
Hub1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
PC-PT
PC6
PC-PT
PC7
192.168.1.1/24
192.168.1.2/24
192.168.1.3/24
192.168.1.4/24
192.168.1.5/24
192.168.1.6/24
192.168.1.7/24
192.168.1.8/24
PC0
Physical Config Desktop Programming Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.8
Pinging 192.168.1.8 with 32 bytes of data:
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time=2ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>
Top
Time: 00:07:58
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Per
Automatically Choose Connection Type

(b)

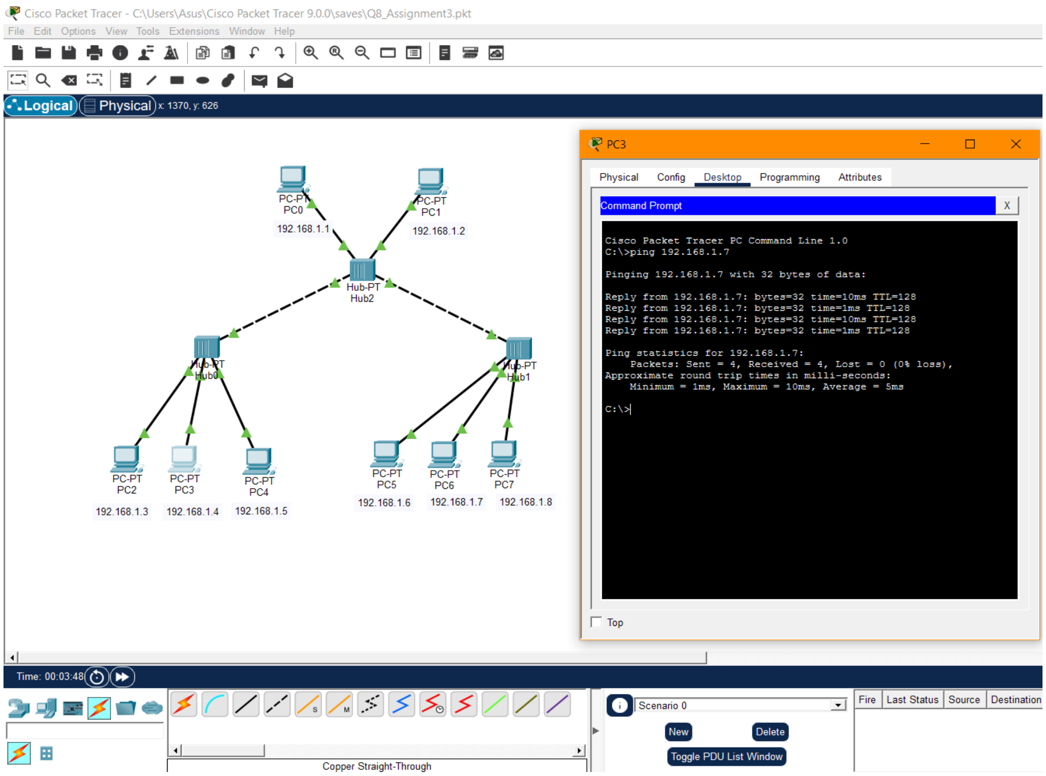
4. Create the following ring topology



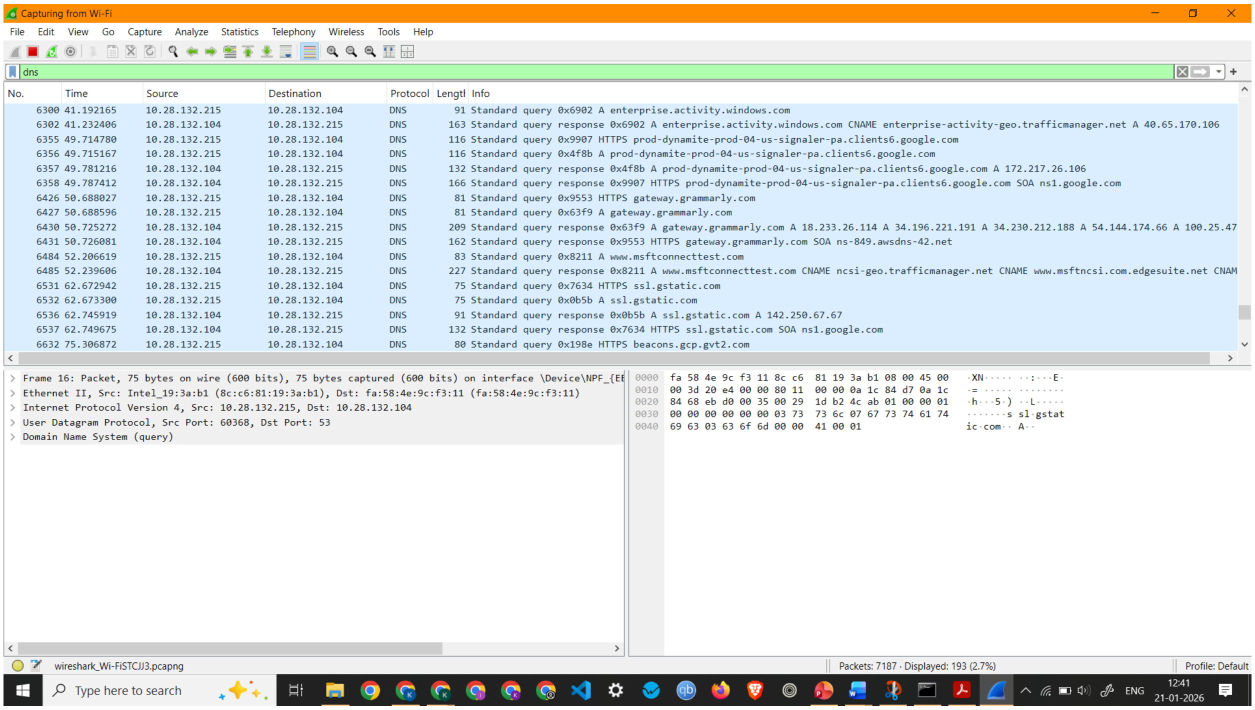
5. Create the following ring topology



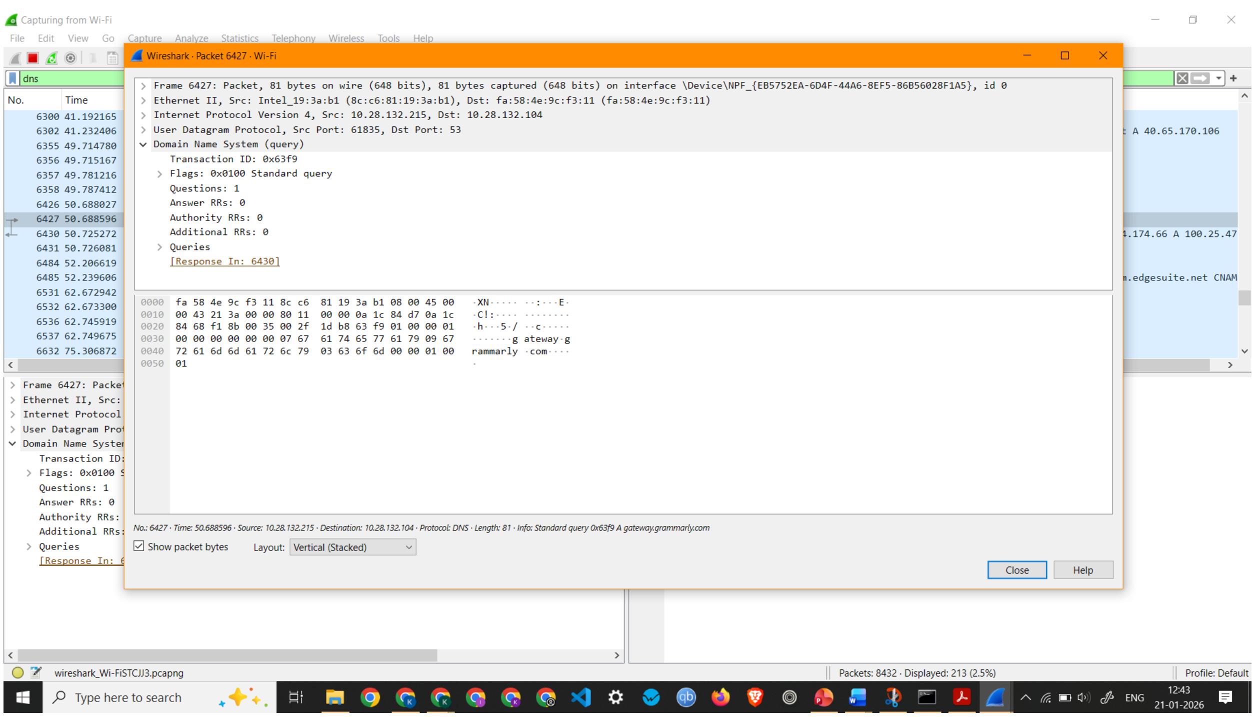
6. Create the following ring topology



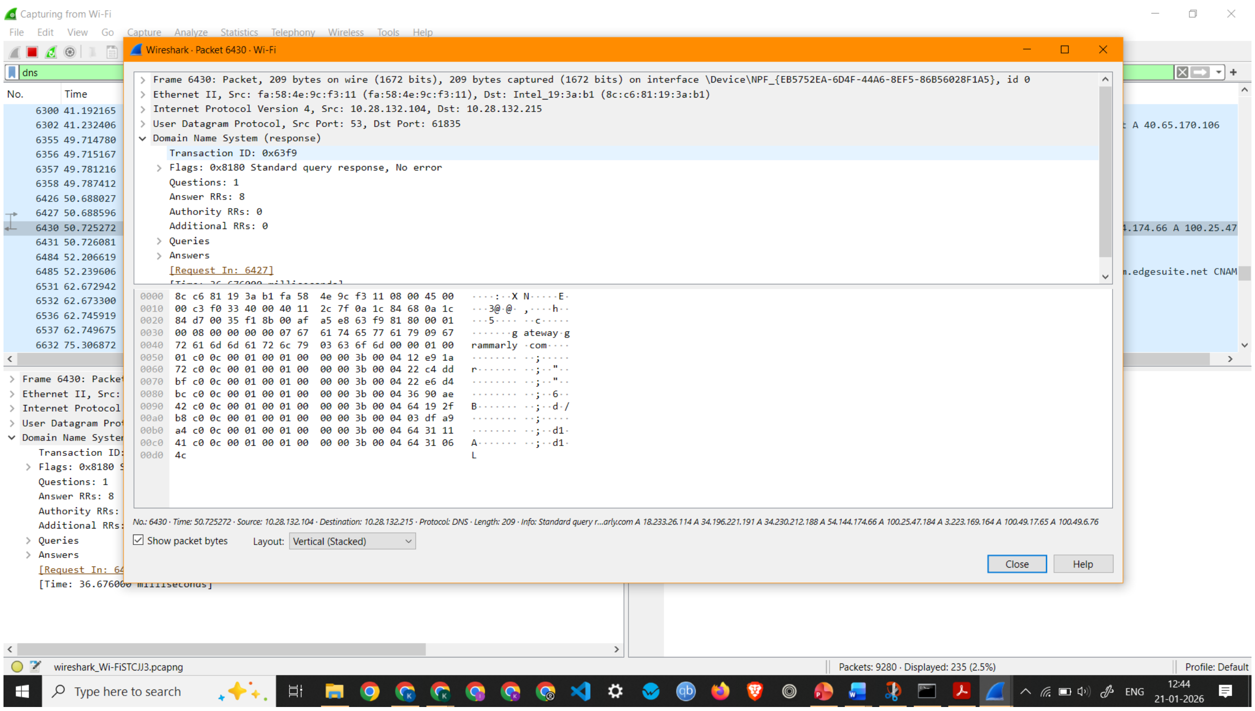
4. To capture and analyze DNS query and response packets



Screenshot showing DNS Packets Capture



Screenshot showing details of a DNS Packet 6427 (Standard query request)



Screenshot showing details of a DNS Packet 6430 (Standard query response)

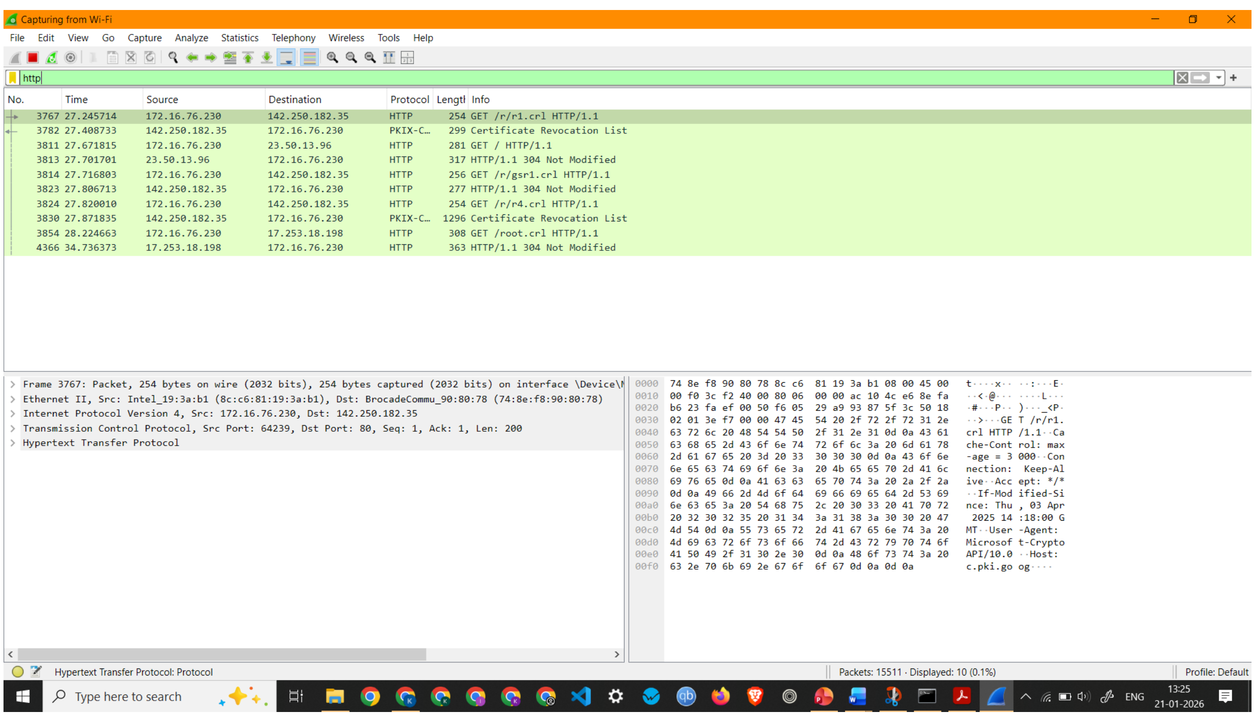
Observation: DNS Packet 6427 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.215
- Destination IP Address: 10.28.132.104
- Packet Length: 81 Bytes
- Queries - gateway.grammarly.com: type A, class IN

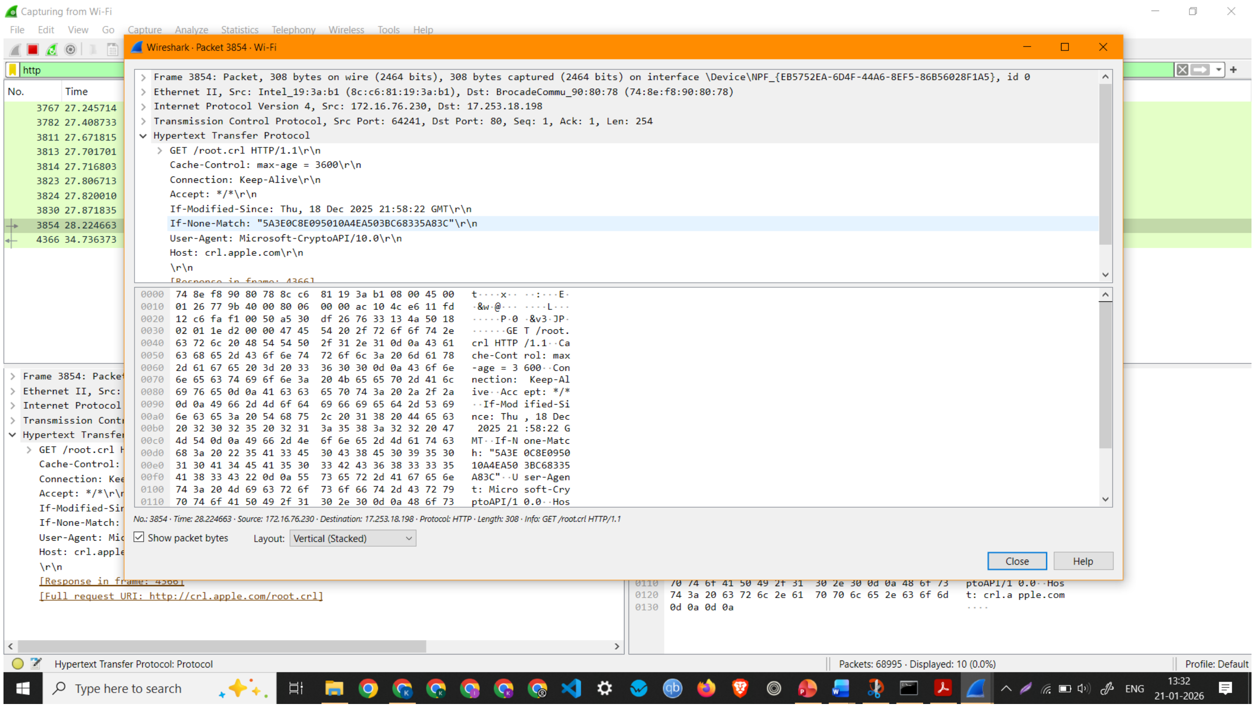
Observation: DNS Packet 6430 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.104
- Destination IP Address: 10.28.132.215
- Packet Length: 209 Bytes

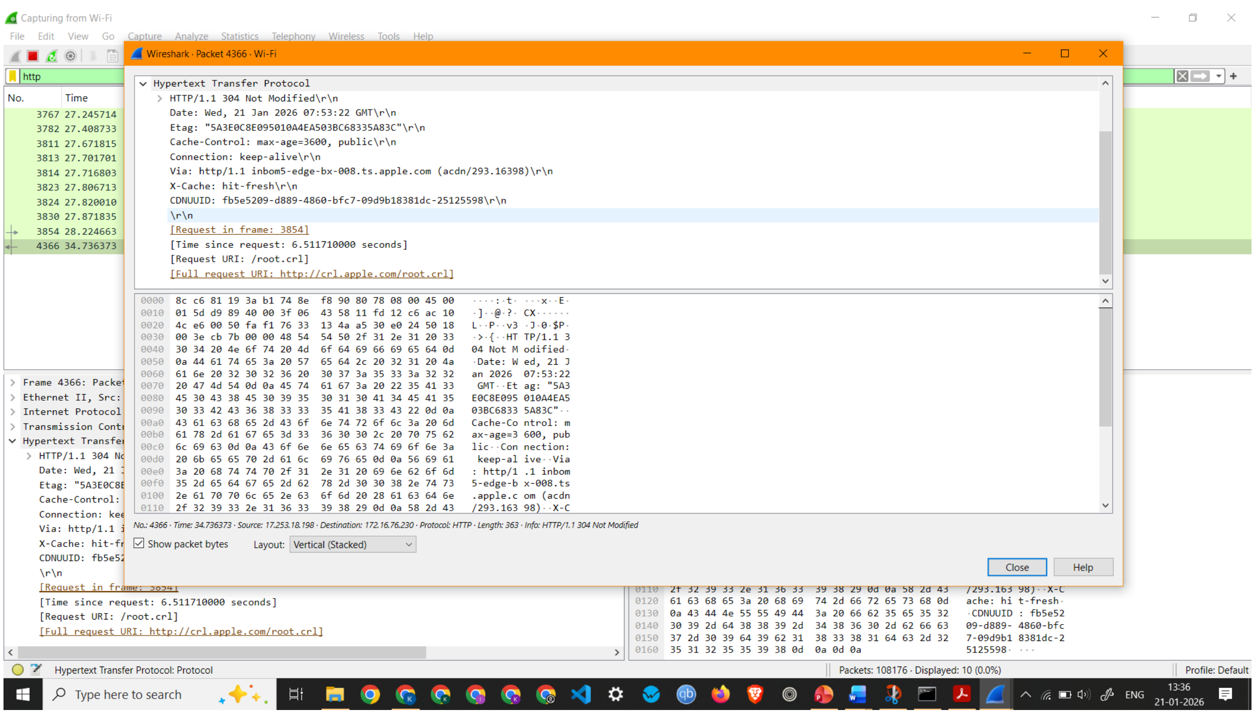
5. To capture and analyze HTTP packets



Screenshot showing HTTP Packets Capture



Screenshot showing details of a HTTP Packet (HTTP Request)



Screenshot showing details of a HTTP Packet (HTTP Response)

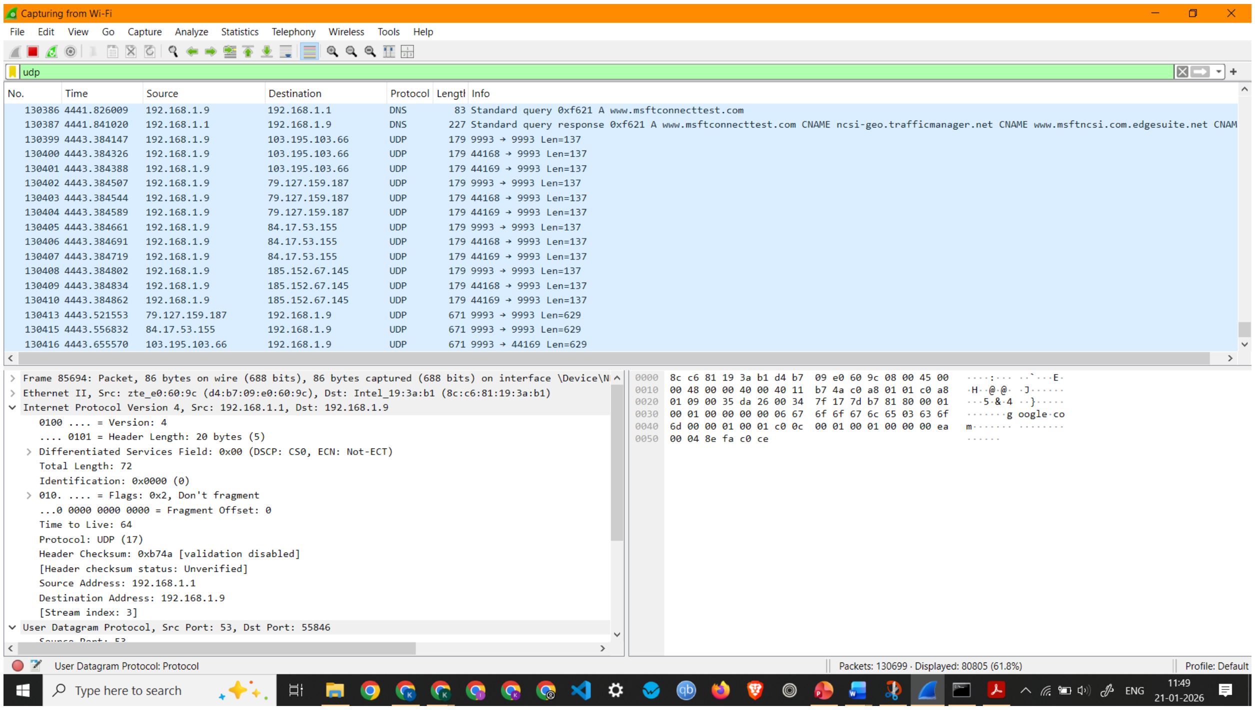
Observation: HTTP Request 3854 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.76.230
- Destination IP Address: 17.253.18.198
- Packet Length: 308 Bytes

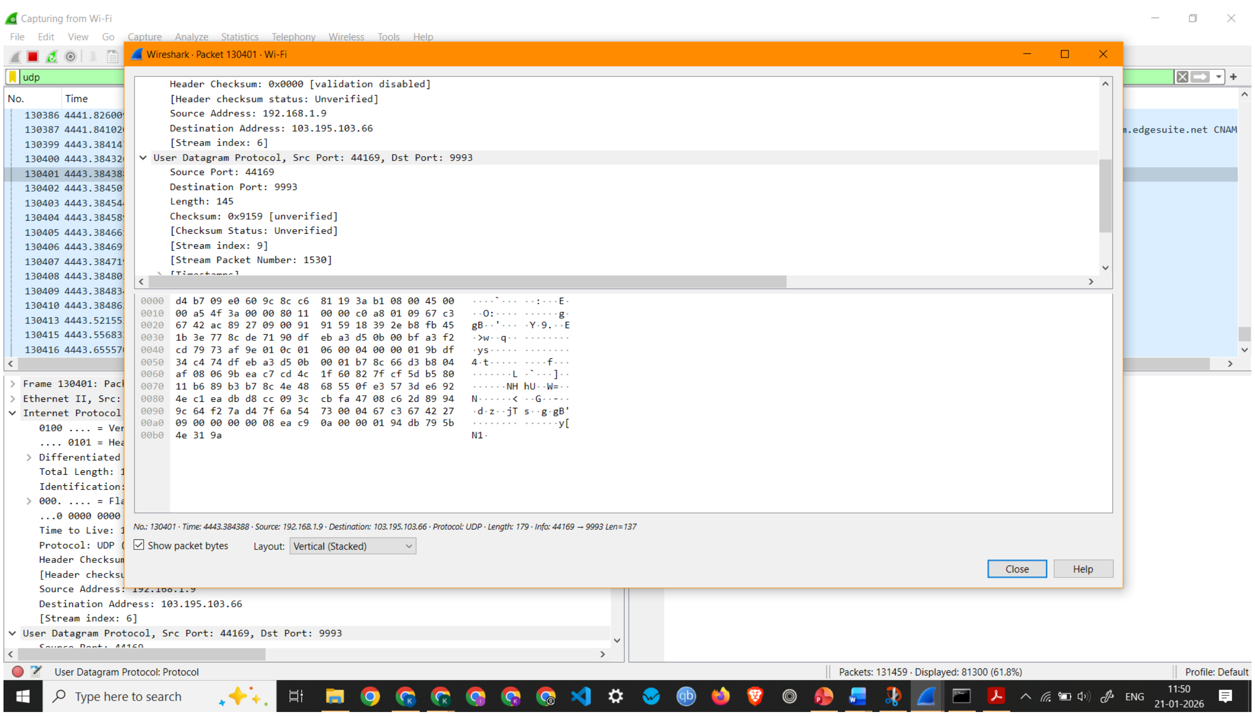
Observation: HTTP Response 4366 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 17.253.18.198
- Destination IP Address: 172.16.76.230
- Packet Length: 363 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

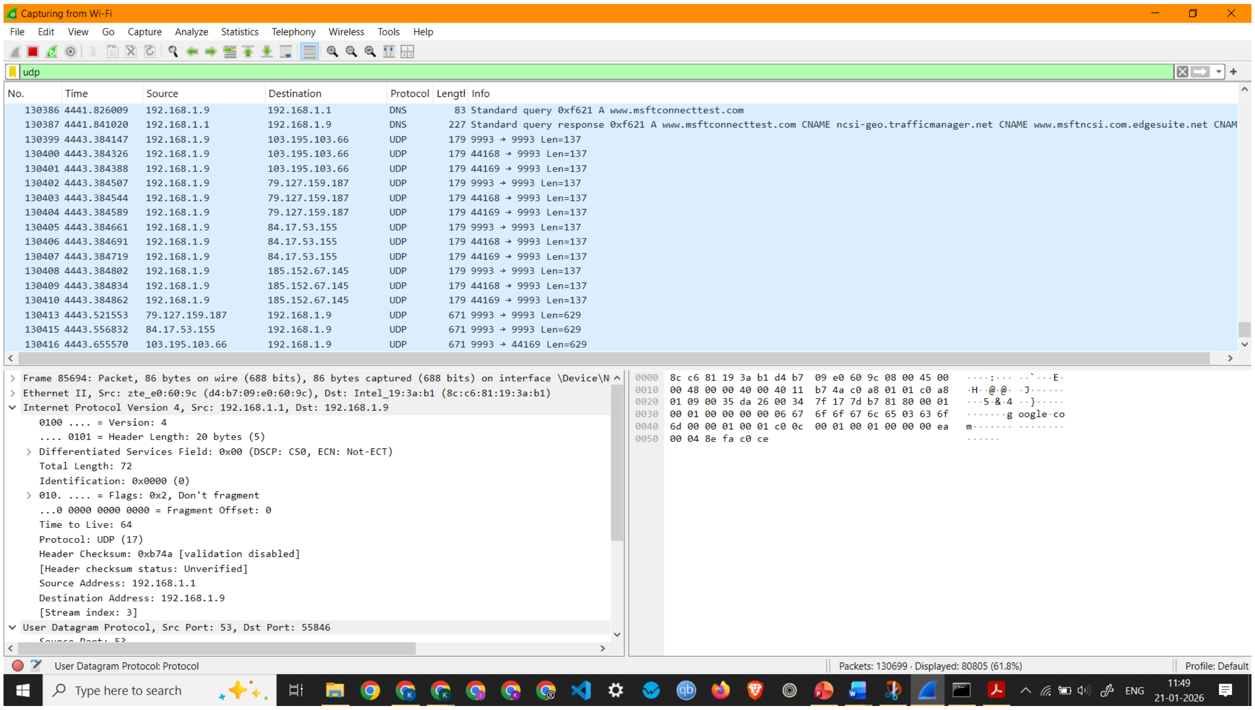


Screenshot showing details of a UDP Packet

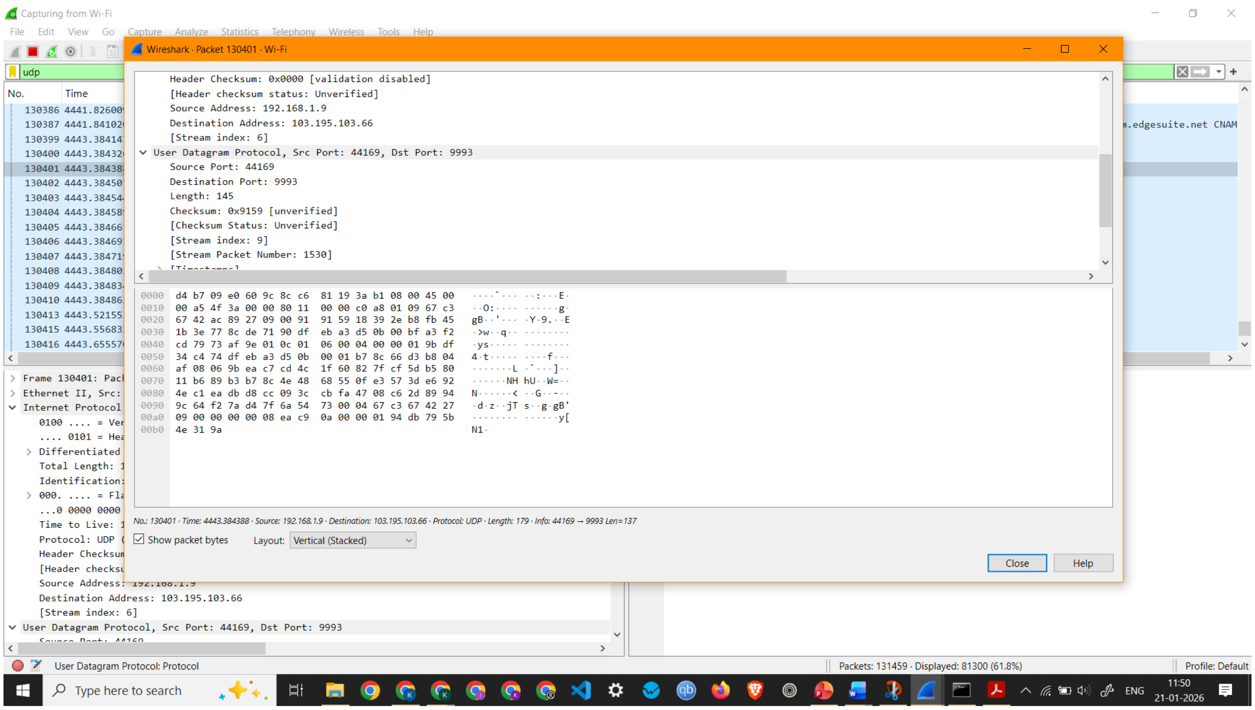
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

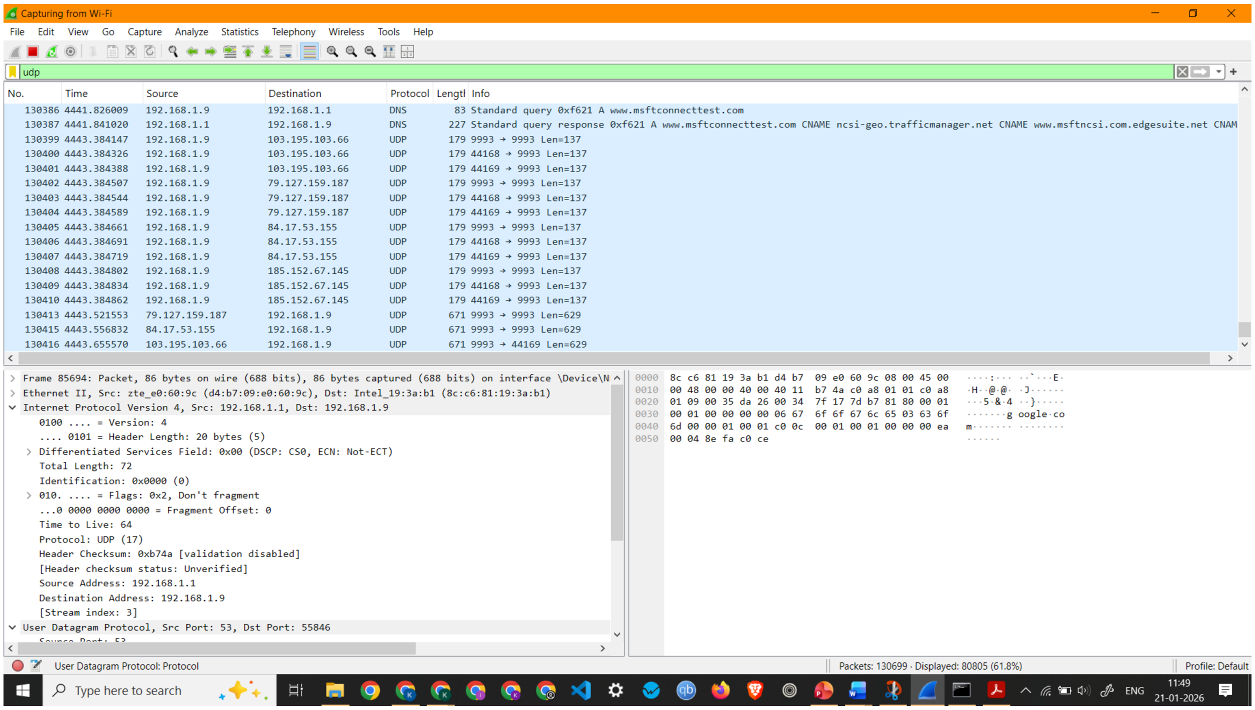


Screenshot showing details of a UDP Packet

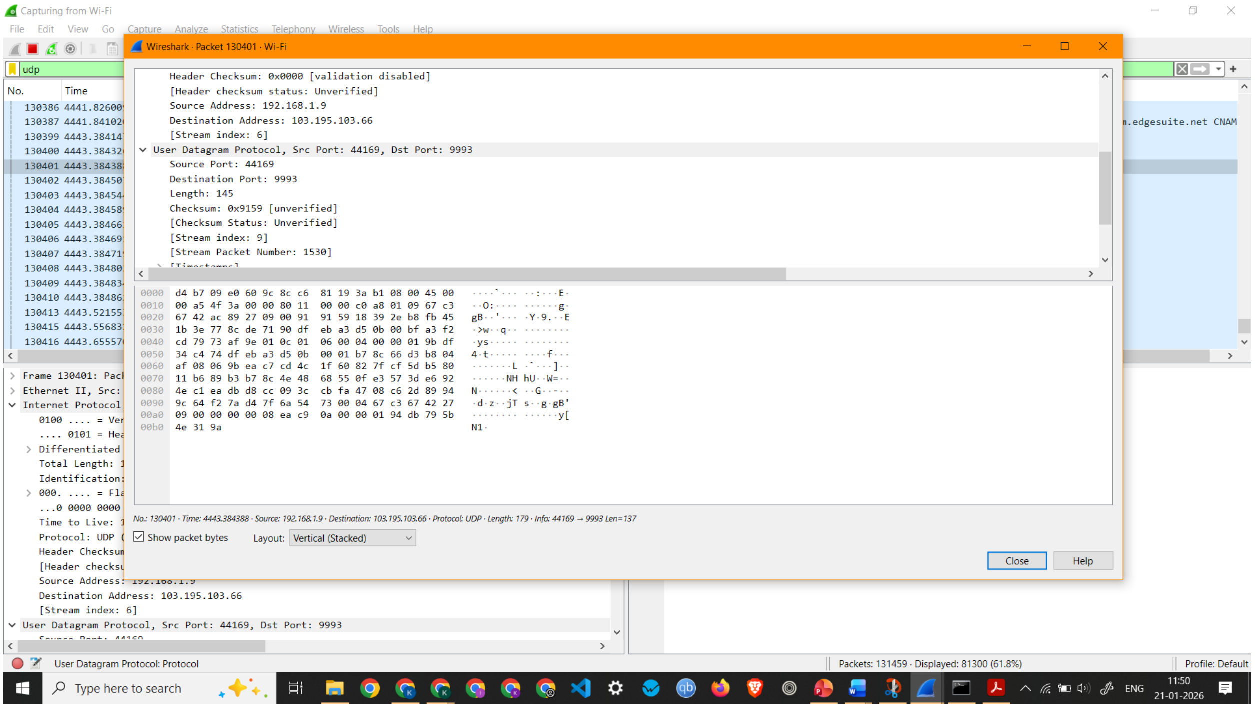
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

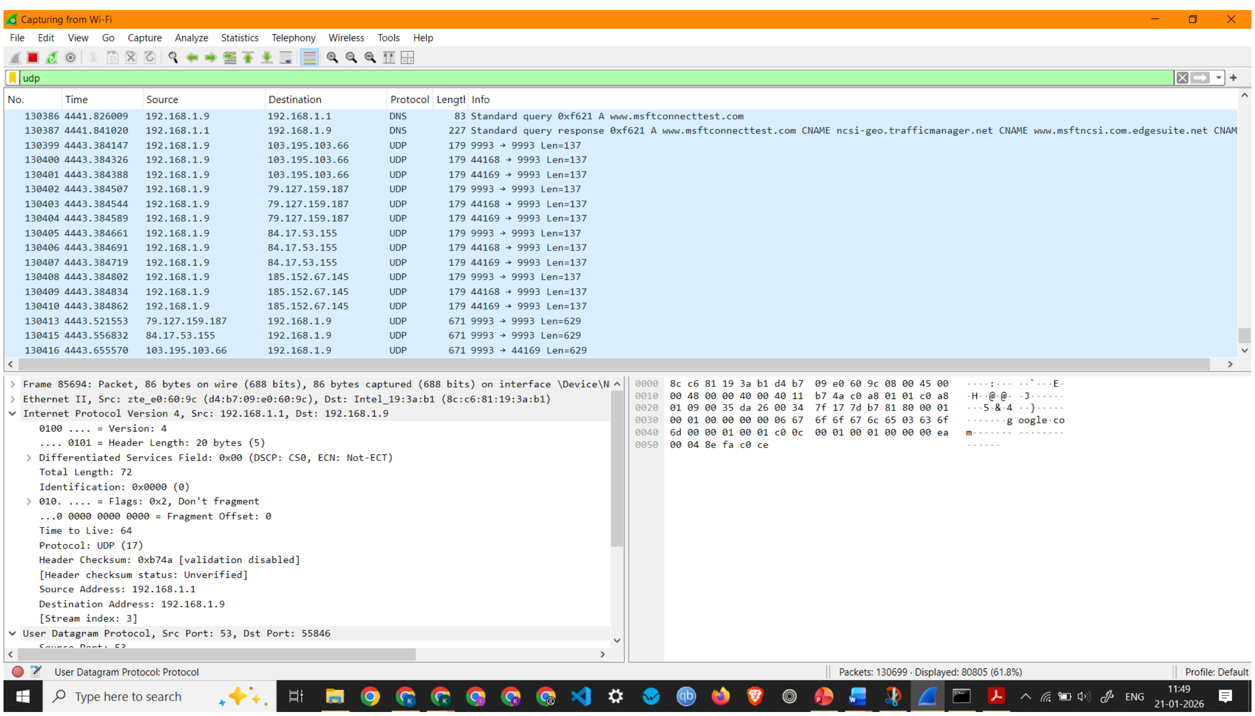


Screenshot showing details of a UDP Packet

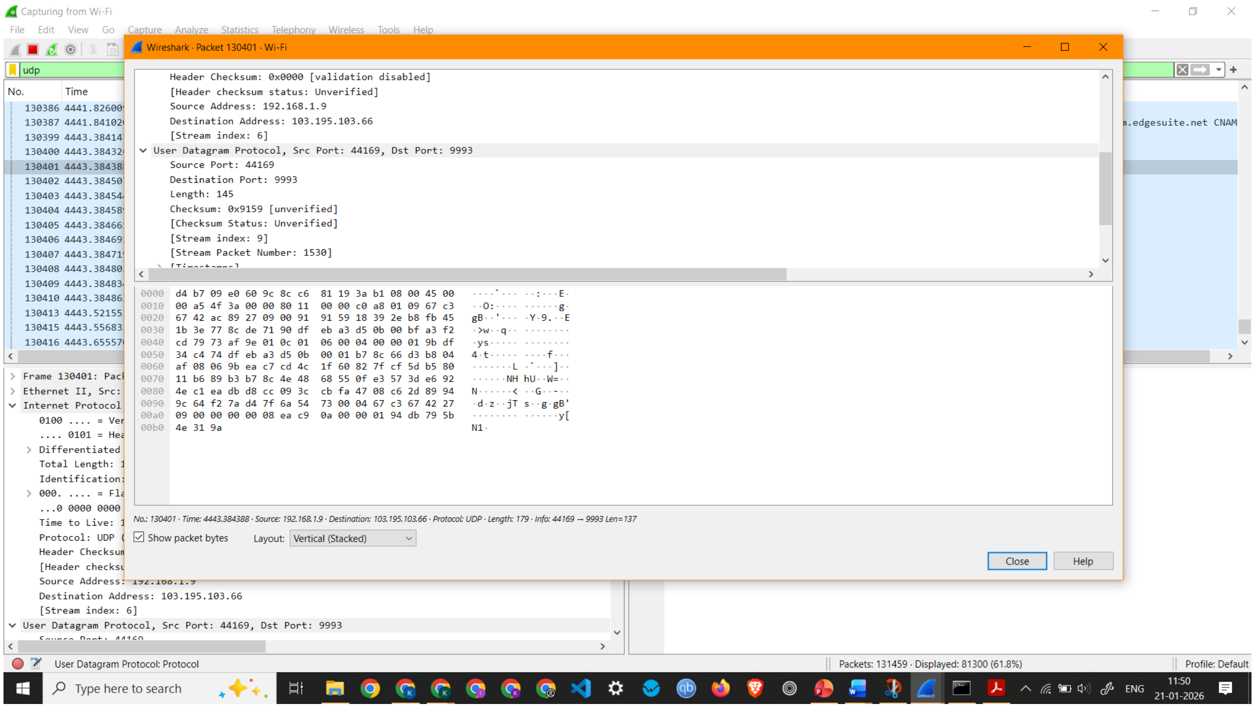
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture



Screenshot showing details of a UDP Packet

Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes